



Apoptosis

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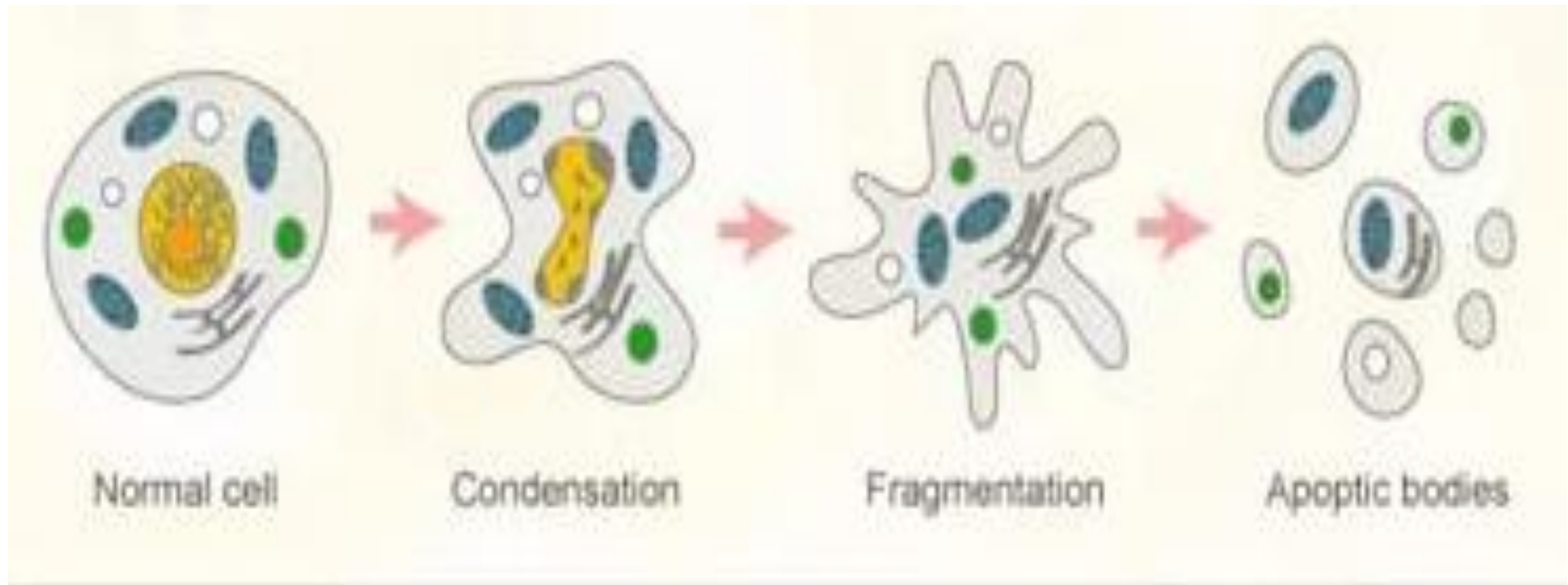
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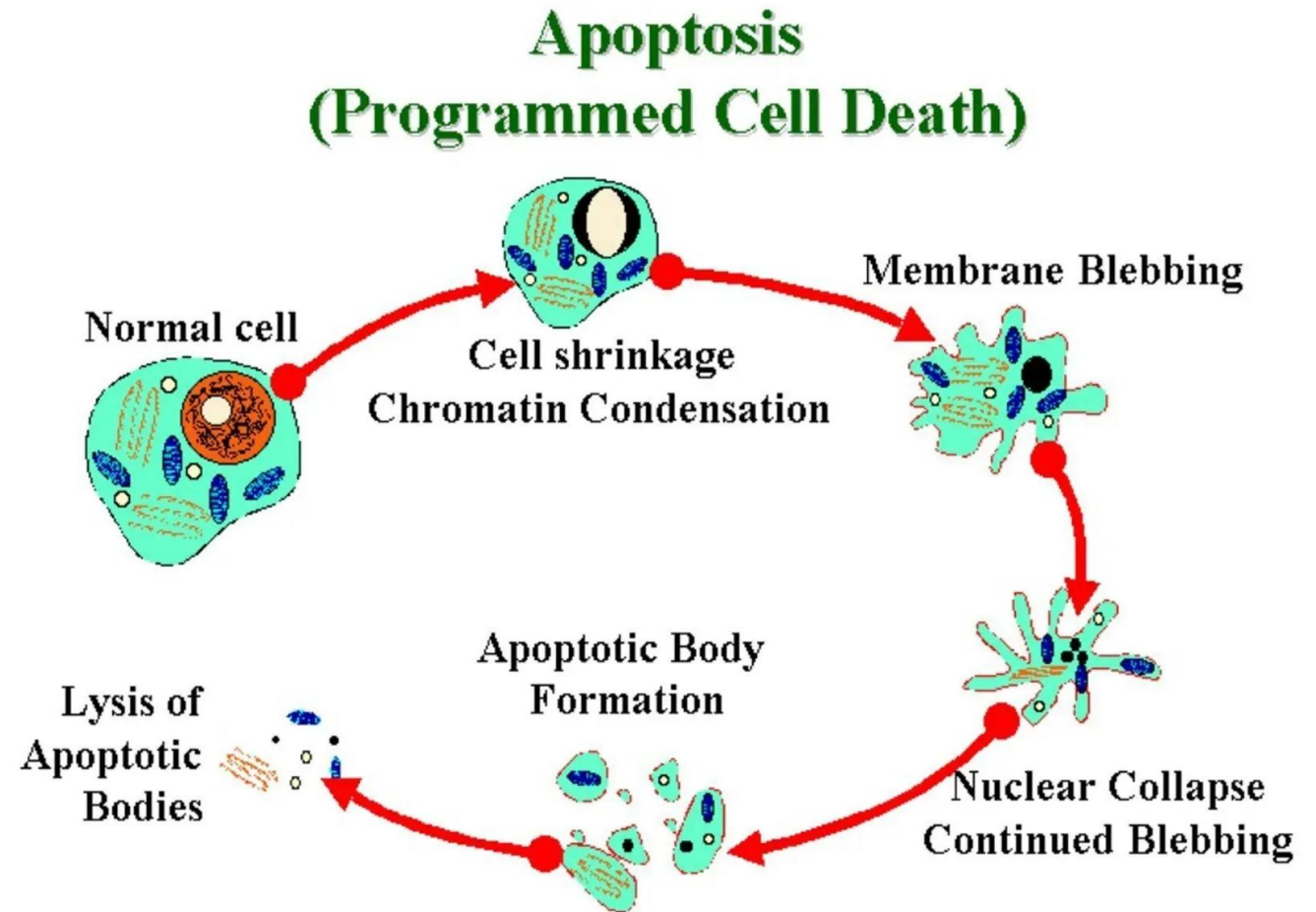
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Apoptosis—Programmed Cell Death

- **Apoptosis** (Programmed cell death) a normal type of cell death that removes unneeded cells during embryological development, regulates the number of cells in tissues, and eliminates many potentially dangerous cells such as cancer cells.

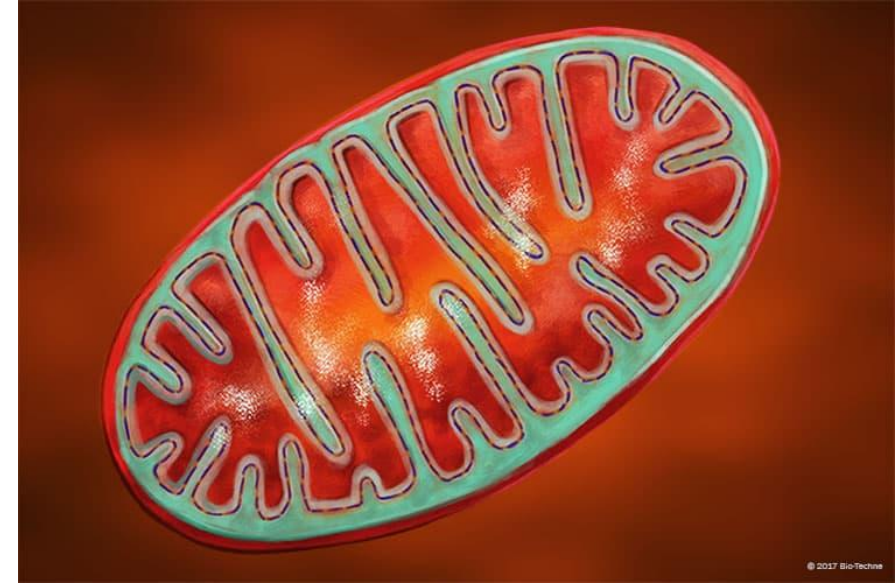


- Apoptosis, is an orderly cell death that results in disassembly and phagocytosis of the cell before any leakage of its contents occurs, and neighboring cells usually remain healthy.



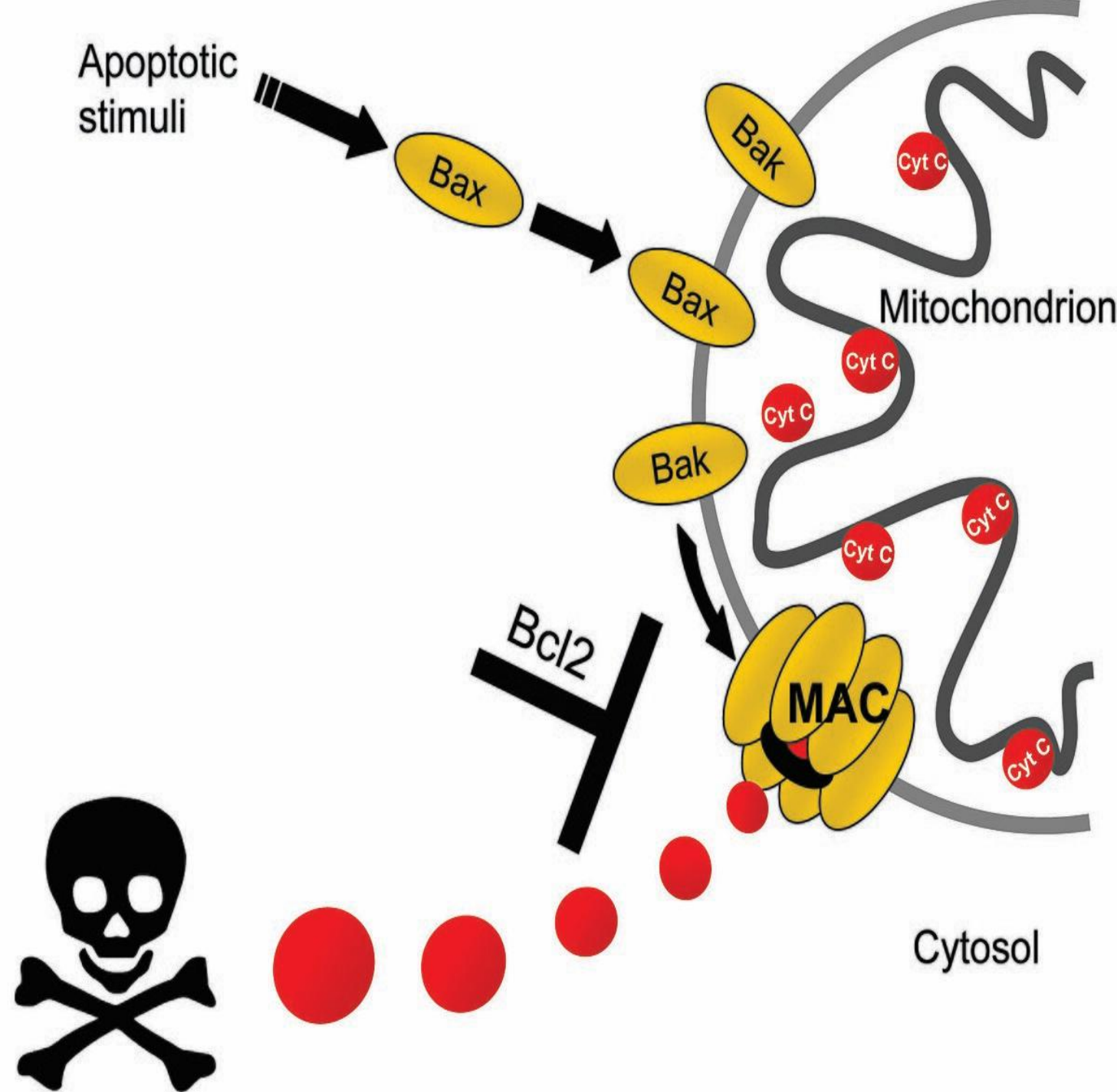
- The cells of the body are members of a highly organized community in which the total number of cells is regulated not only by controlling the rate of cell division but also by controlling the rate of cell death.
- When cells are no longer needed or become a threat to the organism, they undergo a suicidal *programmed cell death*, or *apoptosis*.

Mitochondria play an important and early role in **apoptosis**

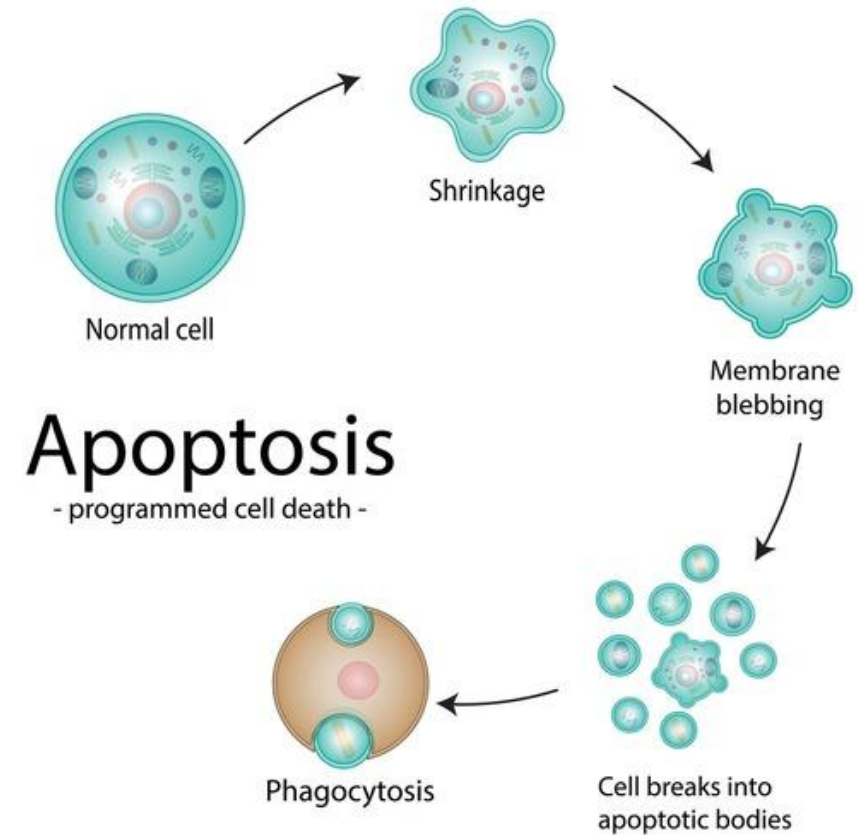


- In response to stimuli such as large numbers of destructive free radicals, DNA damage, growth factor deprivation, or lack of oxygen and nutrients, certain chemicals are released from mitochondria following the formation of a pore in the outer mitochondrial membrane.

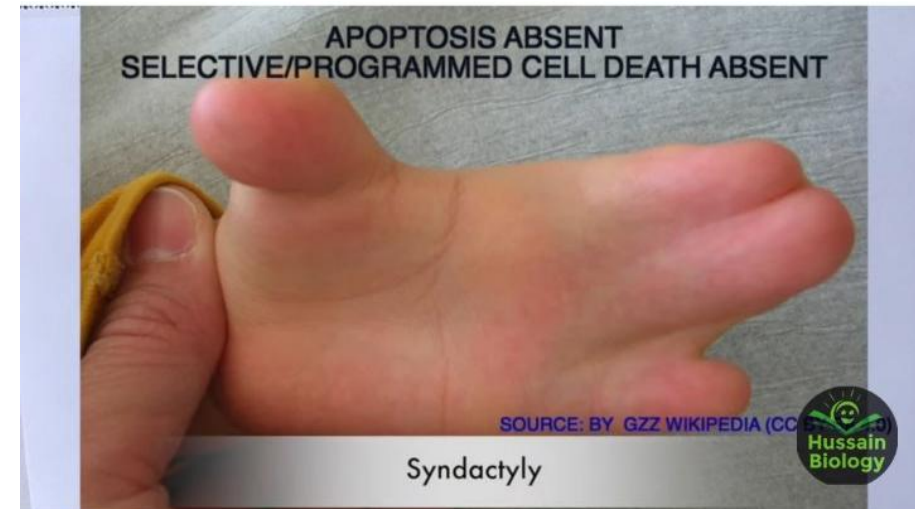
- One of the chemicals released into the cytosol of the cell is **cytochrome *c***, which while inside the mitochondria is involved in aerobic cellular respiration. In the cytosol, however, cytochrome *c* and other substances initiate a cascade of activation of protein-digesting enzymes that bring about apoptosis.



- This process involves a specific proteolytic cascade that causes the cell to shrink and condense, to disassemble its cytoskeleton, and to alter its cell surface so that a neighboring phagocytic cell, such as a macrophage, can attach to the cell membrane and digest the cell.



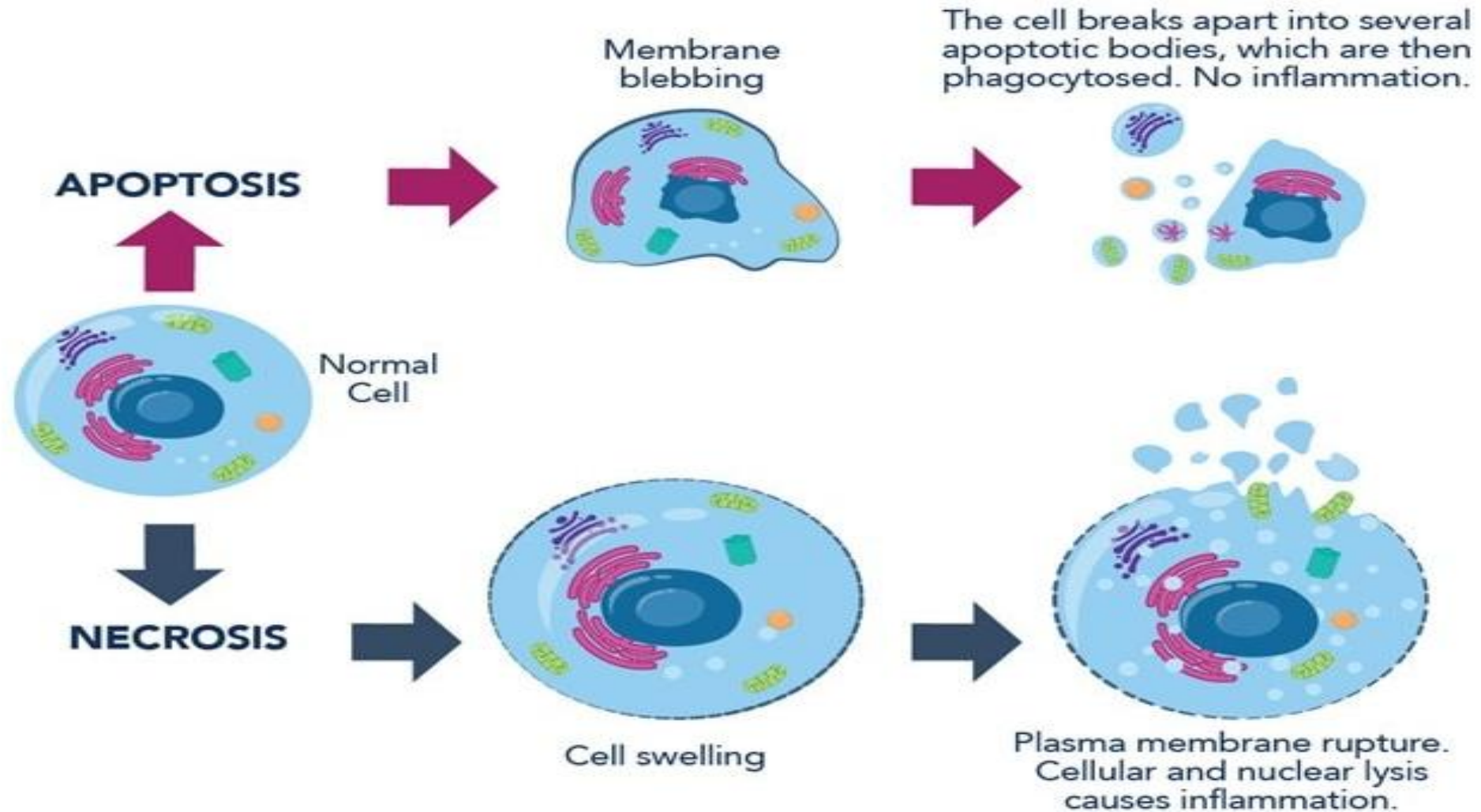
- Nearby phagocytes then ingest the dying cell via a complex process that involves a receptor protein in the plasma membrane of the phagocyte that binds to a lipid in the plasma membrane of the dying cell.
- Apoptosis removes unneeded cells during fetal development, such as the webbing between digits. It continues to occur after birth to regulate the number of cells in a tissue and eliminate potentially dangerous cells such as cancer cells.



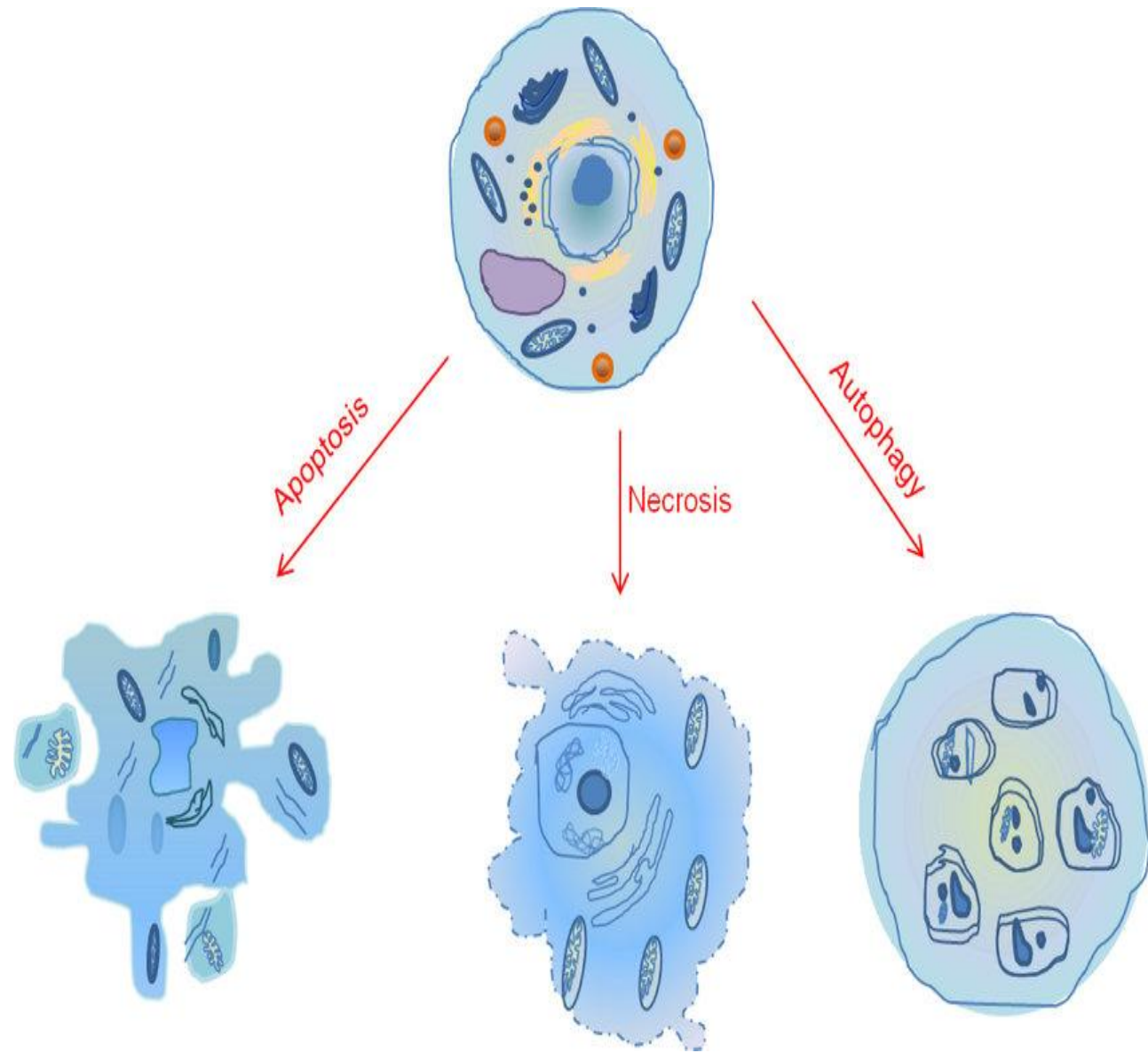
- A tremendous amount of apoptosis occurs in tissues that are being remodeled during development.
- Even in adult humans, billions of cells die each hour in tissues such as the intestine and bone marrow and are replaced by new cells.
- Programmed cell death, however, is normally balanced with the formation of new cells in healthy adults. Otherwise, the body's tissues would shrink or grow excessively.

- Recent studies suggest that abnormalities of apoptosis may play a key role in neurodegenerative diseases such as Alzheimer's disease, as well as in cancer and autoimmune disorders.
- Some drugs that have been used successfully for chemotherapy appear to induce apoptosis in cancer cells.

- In contrast to programmed death, cells that die as a result of an acute injury usually swell and burst due to loss of cell membrane integrity, a process called **cell necrosis**.
- Necrotic cells may spill their contents, causing inflammation and injury to neighboring cells.

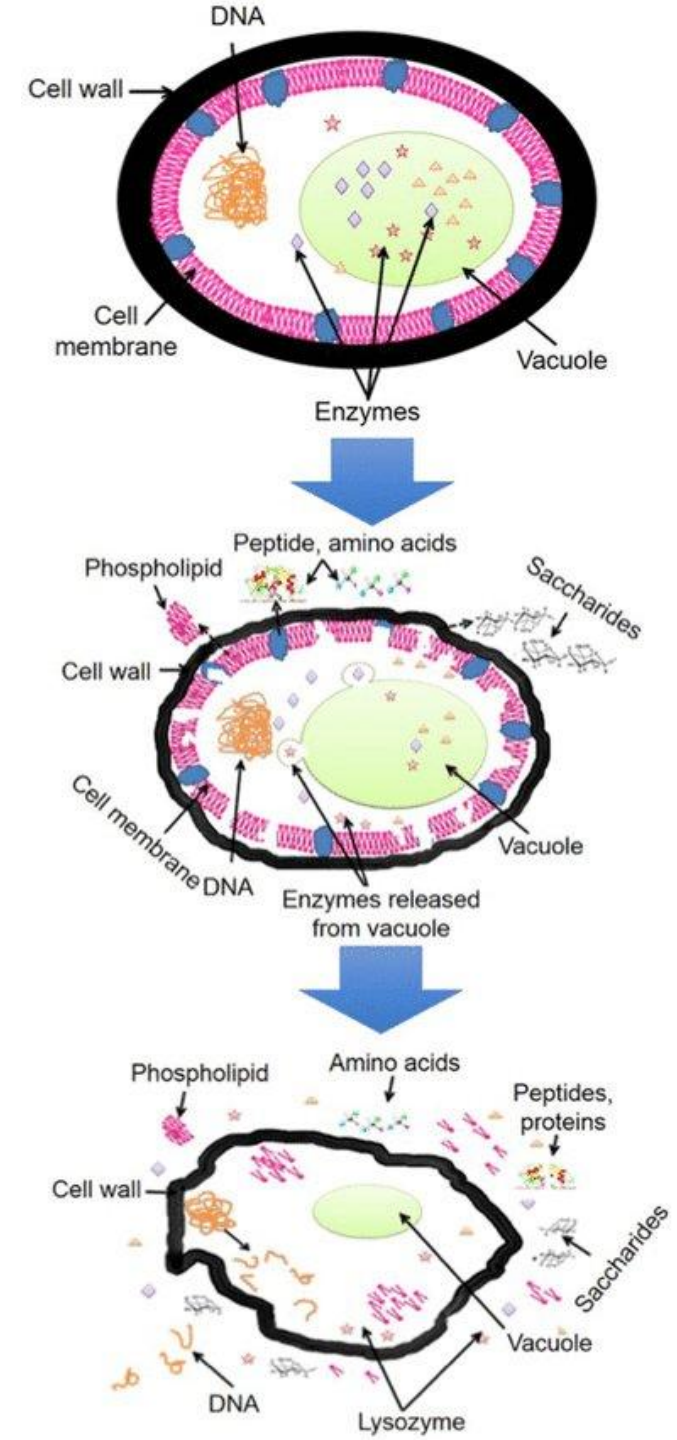


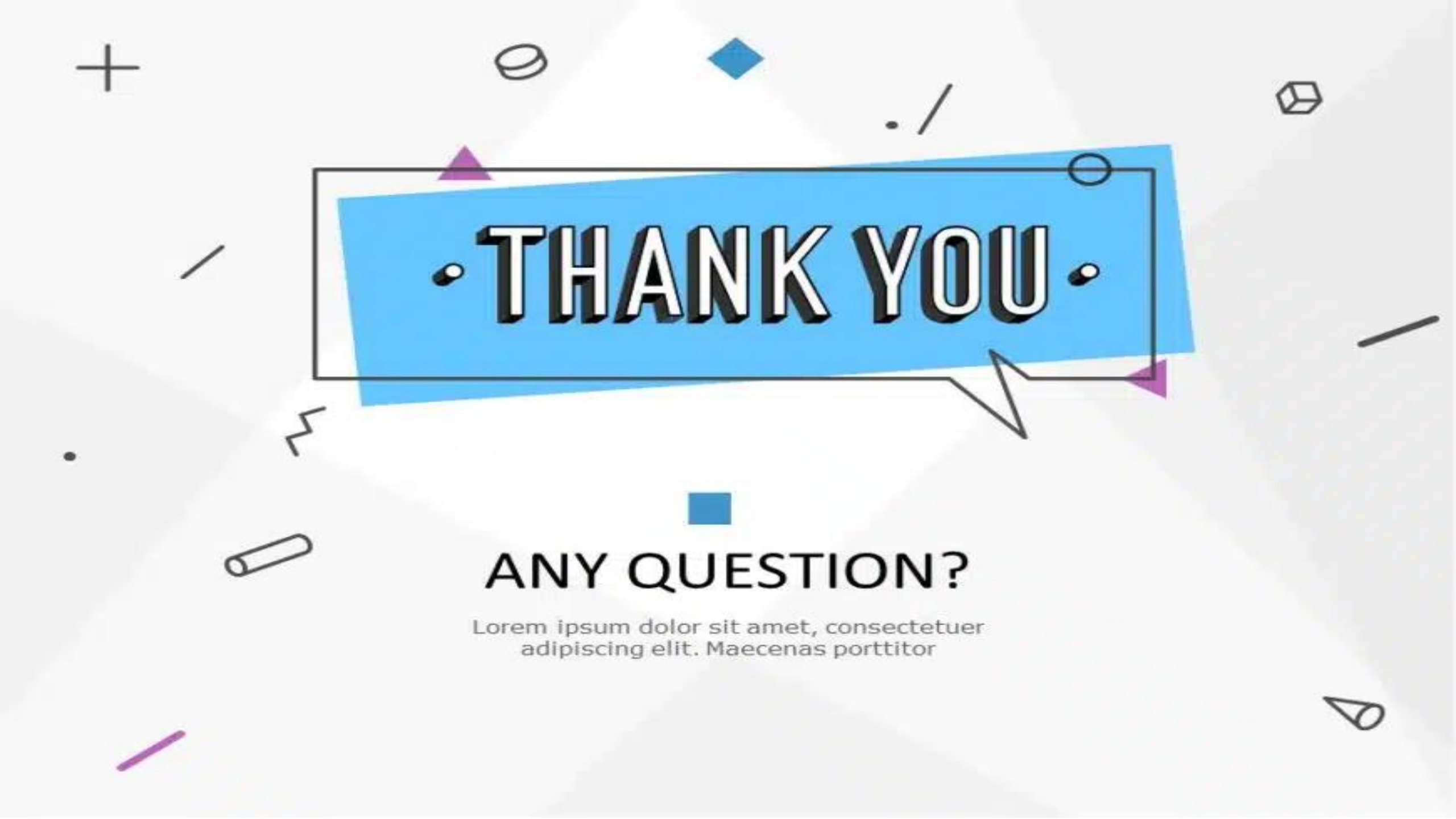
- The process by which entire worn-out organelles are digested is called **autophagy**.
- In autophagy, the organelle to be digested is enclosed by a membrane derived from the ER to create a vesicle called an **autophagosome**.
- the vesicle then fuses with a lysosome. In this way, a human liver cell, for example, recycles about half of its cytoplasmic contents every week.



Autolysis:

1. In this process cell eats itself.
2. It occurs due to injury or infection.
3. It generally not happen in healthy tissues.
4. It is a biological process of cell self destruction.
5. In this process lysosomal enzymes are responsible for cell destruction



The background is a light gray with a subtle geometric pattern of overlapping triangles. Scattered throughout are various hand-drawn style icons: a plus sign, a circle with a dot, a diamond, a cube, a speech bubble tail, a zigzag line, a cylinder, a pencil, and a triangle. A blue rectangular box with a black border and a speech bubble tail on the right side is centered in the upper half of the image.

THANK YOU



ANY QUESTION?

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